4. Construct a Scheduling Program with C that Selects the Waiting Process with the Smallest Execution Time to Execute Next (SJF)

Code - Part 1 (Lines 1–35)

```c
1  #include <stdio.h>
2
3  int main()
4  {
5      int bt[20], p[20], wt[20], tat[20];
6      int i, j, n, total = 0, pos, temp;
7      float avg_wt, avg_tat;
8
9      printf("Enter number of processes: ");
10     scanf("%d", &n);
11
12     printf("\nEnter Burst Time:\n");
13     for(i = 0; i < n; i++)
14     {
15         printf("P%d: ", i + 1);
16         scanf("%d", &bt[i]);
17         p[i] = i + 1;
18     }
19
20     // Sorting based on burst time
21     for(i = 0; i < n; i++)
22     {
23         pos = i;
24         for(j = i + 1; j < n; j++)
25         {
26             if(bt[j] < bt[pos])
27                 pos = j;
28         }
29
30         temp = bt[i];
31         bt[i] = bt[pos];
32         bt[pos] = temp;
33
34         temp = p[i];
35         p[i] = p[pos];
```

Code - Part 2 (Lines 36–70)

```
36          p[pos] = temp;
37      }
38
39      // Waiting time
40      wt[0] = 0;
41      for(i = 1; i < n; i++)
42      {
43          wt[i] = 0;
44          for(j = 0; j < i; j++)
45              wt[i] += bt[j];
46
47          total += wt[i];
48      }
49
50      avg_wt = (float)total / n;
51      total = 0;
52
53      // Turnaround time
54      printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
55      for(i = 0; i < n; i++)
56      {
57          tat[i] = bt[i] + wt[i];
58          total += tat[i];
59
60          printf("P%d\t\t%d\t\t%d\t\t%d\n",
61                  p[i], bt[i], wt[i], tat[i]);
62      }
63
64      avg_tat = (float)total / n;
65
66      printf("\nAverage Waiting Time = %.2f", avg_wt);
67      printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
68
69      return 0;
70  }
```

Output

```
Enter number of processes: 3

Enter Burst Time:
P1: 5
P2: 6
P3: 3

Process Burst Time      Waiting Time    Turnaround Time
P3              3               0               3
P1              5               3               8
P2              6               8               14

Average Waiting Time = 3.67
Average Turnaround Time = 8.33


...Program finished with exit code 0
Press ENTER to exit console.
```